

π -Like Constants For Regular Polygons, And Their Infinite Limit To Circles

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1 Introduction

Well known formulas exist to calculate the perimeter and area of regular polygons from their apothem, (the radius of the incircle they are circumscribed about). Here, I construct a particular π -like constant, k , which can be defined for any n-sided regular polygon circumscribed about a circle of radius r .

$$k = n \cdot \tan\left(\frac{\pi}{n}\right)$$

The exact area of the polygon can be calculated from the equation kr^2 and its perimeter from $2kr$; much akin to the circle's π constant within the $2\pi r$ or πr^2 formulas. The extrapolation from an infinitely sided regular polygon to a circle will also be described.

2 Deriving The Value Of k

The specific polygon-constant, k , may be independently derived from either the perimeter or area calculation of an arbitrary polygon with n sides, circumscribed around a circle of radius r . Since all regular polygons are rotationally symmetric and dissectible into congruent triangles, points A, B, C, D, and angle θ can represent the necessary portion of any n-sided polygon.

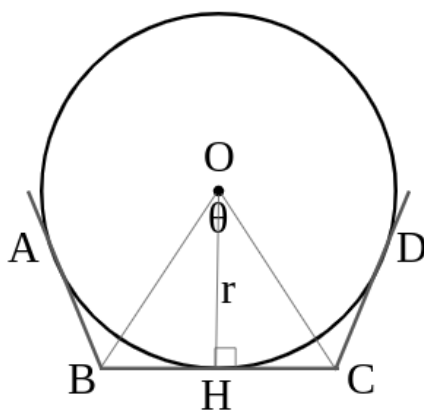


Figure 1: An Arbitrary Polygon ...ABCD... Is Circumscribed About Circle O

2.1 Perimeter Method

$$Perimeter = BC \cdot n$$

Triangle CBO is an isosceles, therefore triangles BHO and CHO are congruent. Thus, angles BOH and HOC are equal to $\frac{\theta}{2}$. Observe:

$$\tan\left(\frac{\theta}{2}\right) = \frac{HC}{r} = \frac{BH}{r}$$

$$Perimeter = 2 \cdot \tan\left(\frac{\theta}{2}\right) \cdot r \cdot n$$

$$Perimeter = 2rn \cdot \tan\left(\frac{\pi}{n}\right)$$

$$Perimeter = 2kr$$

And as might be expected of this hypothetical constant, the radius, r , cancels in the definition.

$$k = n \cdot \tan\left(\frac{\pi}{n}\right)$$

2.2 Area Method

A similar approach can be taken in regards to the area of the circumscribed polygon:

$$Area = n \cdot \frac{BC \cdot OH}{2}$$

Where OH (the radius r) represents the perpendicular height to triangle BOC.

$$Area = n \cdot \frac{BC \cdot r}{2}$$

$$\text{Since } \tan\left(\frac{\theta}{2}\right) = \frac{HC}{r} = \frac{BH}{r}, BC = 2r \cdot \tan\left(\frac{\theta}{2}\right)$$

$$Area = n \cdot \frac{2r \cdot \tan\left(\frac{2\pi}{n}\right) \cdot r}{2}$$

$$Area = nr^2 \cdot \tan\left(\frac{\pi}{n}\right)$$

$$Area = kr^2$$

$$kr^2 = nr^2 \cdot \tan\left(\frac{\pi}{n}\right)$$

Once more, the r -terms cancel, and the area-derived k expression is equivalent to the perimeter-derived expression, above.

$$k = n \cdot \tan\left(\frac{\pi}{n}\right)$$

3 The Limit To Infinity

The properties of a circle may be emulated by an infinitely-sided polygon. As such, k can be evaluated as follows:

$$\lim_{n \rightarrow \infty} n \cdot \tan\left(\frac{\pi}{n}\right)$$

Applying L'Hôpital's Rule:

$$\lim_{n \rightarrow \infty} \left(\frac{\frac{d}{dx} [\tan(\frac{\pi}{n})]}{\frac{d}{dx} [\frac{1}{n}]} \right)$$

$$\lim_{n \rightarrow \infty} \left(\frac{\frac{-\pi \sec^2(\frac{\pi}{n})}{n^2}}{\frac{1}{-n^2}} \right)$$

$$\begin{aligned}
& \lim_{n \rightarrow \infty} \left(\pi \sec^2 \left(\frac{\pi}{x} \right) \right) \\
& \pi \sec^2 \left(\lim_{n \rightarrow \infty} \left(\frac{\pi}{n} \right) \right) \\
& \pi \sec^2 (0) = \pi
\end{aligned}$$

Therefore, for an infinitely-sided polygon, $k = \pi$ (a graphic depiction is shown in Figure 2). The area of this shape, predictably, is then πr^2 and its perimeter, or circumference rather, is $2\pi r$.

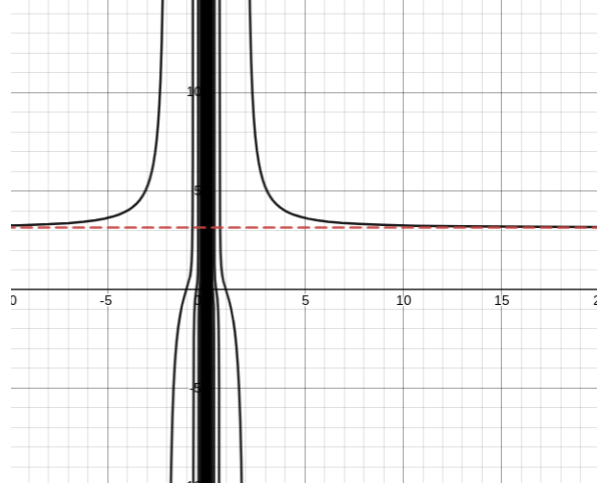


Figure 2: The Function $y = x \cdot \tan \left(\frac{\pi}{x} \right)$ Approaches The Horizontal Asymptote of $y = \pi$

3.1 Proof Of Circle Area And Circumference

As shown in Section 2, the perimeter and area of an n-sided polygon is equal to:

$$Perimeter = 2rn \tan \left(\frac{\pi}{n} \right)$$

$$Area = nr^2 \tan \left(\frac{\pi}{n} \right)$$

These can each be expressed in terms of k , and evaluated as n tends to positive infinity.

$$\lim_{n \rightarrow \infty} (2rk)$$

$$\lim_{n \rightarrow \infty} (2rk)$$

The Limit Product Law can be applied:

$$Perimeter = \lim_{n \rightarrow \infty} (2r) \cdot \lim_{n \rightarrow \infty} (k)$$

$$Area = \lim_{n \rightarrow \infty} (r^2) \cdot \lim_{n \rightarrow \infty} (k)$$

$$Circumference = 2\pi r$$

$$Area = \pi r^2$$

4 Appendix

1. From Section 2.1, length BC is equal to:

$$BC = 2 \cdot \tan \left(\frac{\pi}{n} \right) \cdot r$$

Thus, the apothem length, r , can be found from the polygon's side-length, BC, by rearrangement:

$$r = \frac{BC}{2 \tan \left(\frac{\pi}{n} \right)}$$

2. The $\tan()$ of $\frac{\pi}{3}, \frac{\pi}{4}, \frac{\pi}{6}$ are defined on the Trigonometric Unit Circle, and their respective k-values are shown below.

Number Of Sides	k Constant
3	$3\sqrt{3}$
4	4
6	$2\sqrt{3}$
...	...

Table 1: Several Unit-Circle Values of $\tan\left(\frac{\pi}{n}\right)$ Are Well Defined